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PERSHING II SIMULATION STUDIES

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PREPARED FOR
Systems Simulation and Development Directorate
US Army Missile Laboratory

July 1982

U.S. ARMY MISSILE COMMAND
Redstone Arsenal, Alabama 35809

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Pershing II flight simulations were performed using the U70 missile simulation program to determine (1) in the event of an accidental nozzle deflection, how fast the missile would leave its (safe) flight corridor; (2) how well the U70 aerodynamic simulation matches the actual flight data; and (3) the trajectory profiles for nine Tactical Ballistic Missile flight trajectories. Also an advanced simulation program developed by TRW, Inc., was partially converted to run on an inhouse computer.

PREFACE

This report was prepared by the Electromagnetics Laboratory of the Engineering Experiment Station, Georgia Institute of Technology, under Contract No. DAAH01-81-D-A003, Delivery Order No. 38, for the U. S. Army Missile Command. The contract technical monitor was M. M. Hallum and the contract project monitor was D. L. Cobb, DRSMI-RDF, both from the Systems Simulation and Development Directorate of the U. S. Army Missile Command. The contract period extended from February 3, 1982, through May 28, 1982.

The authors would like to acknowledge the valuable contributions of J. P. Billingsley, and D. L. Cobb.

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I. INTRODUCTION

A. Statement of the Task and General Approach

This report documents the results of the work performed under the Pershing II Task Order DAAH01-81-D-A003, Delivery Order 0038 (GT/EES Project No. A-3165). The task order required that the following subtasks be performed:

- The range safety problems which would arise if the motor nozzle were unintentionally deflected during boost phase were to be evaluated.

- The aerodynamic simulation model used in a Pershing II computer-simulation program was to be validated against actual flight test data.

- Computer simulation results were to be compared with actual flight data for a set of Tactical Ballistic Missile (TBM) trajectory profiles.

The results of these comparisons were "to be presented in a form useful for evaluation".

B. Background

The task documented with this report was an outgrowth of an earlier task (Pershing II Debris Studies, DAAH01-81-D-A003, Delivery Order 0017, GT/EES Project No. A-2946). The earlier task was aimed at evaluating launch and target sites for Pershing II firings, determining the effect of varying aerodynamic conditions on the missile's flight behavior, and conducting range safety studies. A follow-on task order request is in preparation and will expand upon the work performed in support of these two projects.

C. Contents of This Report

A detailed discussion of the subtask objectives and accomplishments is given in Section II. Section III discusses the U70 and TRW simulation programs used to carry out Pershing II missile simulations [1]. It includes a description of the U70 program and the computer on which it runs, modifications that had to be made to it to support this task, and the operational procedures for running it. It also discusses the problems involved in converting the TRW program to run on the U. S. Army Missile Command's Perkin-Elmer 3220 computer. Section IV discusses the results that were obtained in the execution of this task. Finally Section V presents recommendations for future work.

II. DETAILED OBJECTIVES

A. Subtask Objectives and Accomplishments

During the performance of this task, two of the three required computer simulation studies were performed, using the U70 trajectory simulation program [1] and the TRAJ targeting program. The third study, the comparison of Pershing II computer simulation studies with actual flight data, could not be carried out because flight test data were not available during the period of performance of the task. The effort that would normally have been expended upon this comparison of flight test data and simulation results was directed instead toward the implementation of a Maximum Likelihood Method computer subroutine for data analysis.

Similar studies were to be undertaken using a new Pershing II simulation program developed by TRW, Inc., but the full computer program could not be obtained in time for use on this task. Its treatment in this report is necessarily restricted to a discussion of the conversion and documentation of the program itself rather than the results obtained using it. A more detailed description of the four subtask objectives is given below.

1. Flight Safety (Motor Nozzle Deflection) Study

The motor nozzle deflection study was a flight safety study conducted to determine the breakup point of the Pershing II missile in the event of an engine malfunction. A nozzle deflection would cause the missile to deviate from its normal flight path. The objective of the study was to determine how quickly the missile would leave its flight corridor in the event of a spurious nozzle deflection and whether Range Safety would detect this aberration in time to permit destruction of the missile before it could endanger life or property.

2. Aerodynamic Model Validation Study

A validation of the aerodynamic mathematical model used in the U70 Pershing II digital simulator was required for utilization of flight data. To compare the simulation results to the flight data, a statistical technique called the Maximum Likelihood Method was to be used. This technique de-serves special mention because of its key role in the performance of this task. This technique is used to fit theoretically predicted values of relevant variables to experimentally derived values when the experimental values contain random measurement errors. Generally, the Maximum Likelihood Method provides better estimates of the underlying true values of variables than a least-squares curve fit because, in making its estimates, the Maximum Likelihood Method uses a prior knowledge of the standard deviations of instrument measurement errors as additional information to help refine its estimates.

3. Tactical Ballistic Missile Trajectory Study

The Tactical Ballistic Missile (TBM) study took the form of a matrix of U70 runs with various flight paths along one dimension and different flight characteristics along the other dimension.

One launch site and three target sites were used to generate the three flight paths. Then for each flight path, a set of three runs was made for three assumed conditions - first, a "nominal motor" or unperturbed-baseline flight plan; second, a ballistic re-entry vehicle flight plan in which no aerodynamic corrections are applied to the missile during its terminal re-entry phase; and third, a flight plan assuming offsets in target latitude and longitude from its normal target location. Thus, a total of nine runs were made for this study.

B. Conversion and Installation of The TRW Simulator

The TRW program is a six-degree-of-freedom simulator developed for the purpose of validating Pershing II onboard software. This simulator contains aerodynamic modeling features and provides simulation capabilities not found in the U70 simulation program - capabilities which the U. S. Army Missile Command's Systems Simulation and Development Directorate, Systems Evaluation Branch, seeks to add to its repertoire of simulation programs. The job of understanding the program and converting it to the Missile Command's computer was assigned to Georgia Tech as a part of this task (A-3165). TRW delivered a copy of its program to the Systems Evaluation Branch. However, when program compilation and link loading was attempted, certain sections of the program, such as the MAIN control program and the INTERP routine were found to be missing. These could not be obtained during the period of performance of the task, and therefore, could not be converted. Insofar as possible, the remainder of the program was partially converted from TRW's Digital Equipment Corporation VAX 11/780 computer to the System Simulation and Development Directorate's Perkin-Elmer (PE) 3220 computer after making minor changes to accommodate for the differences between the FORTRAN capabilities of the two computers.

Although the installation of the TRW program on the PE 3220 computer and the incorporation of its aerodynamic model within the U70 program is not explicitly required in this task's scope of work, it is implicitly required for the proper performance of the task. Since it has consumed much of the task's effort, it will be discussed in this report as though it were a fourth task objective.

III. THE U70 AND TRW SIMULATION PROGRAMS

A. The U70 Program

1. Description of the U70 Program

The U70 trajectory program employed in the performance of this task, uses the given prelaunch, target, and flight conditions to simulate the flight of a single-stage or two-stage missile and its associated maneuvering re-entry vehicle. This program utilizes the equations and requirements found in the Pershing Launch Computer and the Pershing Airborne Computer to simulate the missile flight from launch through the boost, midcourse, and terminal portions of flight to impact [2].

The U70 trajectory program runs on a Perkin-Elmer 3220 computer system equipped with 800 kilobytes of random access memory, a nine-track tape drive, three 67-megabyte disk drives, and five demand terminals (including the system console). The simulation is coded in FORTRAN VII-D, although it is probably compatible with other versions of FORTRAN.

The program is able to restart the missile simulation at any time during the terminal re-entry portion of flight upon entry of data such as time, position, velocity, and orientation of the missile at the desired restart time, plus the restart data acquired from a previous simulation.

2. Modifications to the U70 Program

a. Modifications for the Motor Nozzle Deflection Study

For this study, modifications were made to the U70 program's BAPLT (Boost Autopilot) subroutine to enable desired deflection angles to be input at some time during boost. These modifications are as follows:

```
     MODIFICATIONS TO BAPLT

470  CONTINUE
     IPND = ICON(50)
     IYND = ICON(51)
     IF(IPND .EQ. 0)  GO TO 453
     IF (TIMC .LT. CIN(382)) GO TO 453
     DNV (1, 1) = CIN (380) * DTOR

453  CONTINUE
     IF(IYND .EQ. 0) GO TO 454
     IF(TIMC .LT. CIN(382)) GO TO 454
     DNV (1, 2) = CIN(381) * DTOR

454  CONTINUE
     RETURN
     END
```

b. Aerodynamic Model Flight Validation

As mentioned previously, the Maximum Likelihood Method (MLM) of data correlation was intended to help fit the simulator-generated aerodynamic flight data to the actual test-flight telemetry data, i.e., to "average out" random telemetry instrumentation errors in order to provide more meaningful comparisons between the simulated and experimental results. The machinery for accomplishing this was incorporated into a new U70 subroutine called DATACORR. The addition of DATACORR to the U70 program required the creation of a new link procedure, TU70LINK.CSS, which builds a new task called TU70.

The TU70 task requires as inputs the instantaneous inertial-frame X(t), Y(t), and Z(t) position coordinates and $\phi(t)$, $\theta(t)$, and $\psi(t)$ (roll, pitch, and yaw) angular coordinates of the missile and their time derivatives (where t represents time). For program checkout purposes, constant offsets have been programmed into the inertial position and rotation inputs. Since the MLM uses an iterative technique, an upper limit must be entered on the number of iterations that the computer is to attempt before giving up (in case the iteration sequence does not converge). Finally, a 6 x 6 matrix of standard deviations and cross-correlations of measurement noise must be entered for the three position and the three rotation coordinates at $t = 0$, e.g., at launch (Table 1).

This matrix contains in its [1, 1] location, the variance (σ_{θ}^{2}) of the instrumentation measurement errors of the missile's x-coordinate at $t = 0$. In its [1, 2] location, it contains the mean square error in X due to an error in Y ($\sigma_{\theta\phi}^{2}$), and so forth, for all 36 elements. These TU70 input requirements are summarized in Table 2. The coding for the DATACORR subroutine is reproduced in Appendix A.

c. Data Format Modifications for the Tactical Ballistic Missile Simulations

Modifications were made for the Tactical Ballistic Missile trajectory simulations to provide the same output format for all stages of flight. Different variables were output for the boost and for the re-entry stages; these variables were combined to produce one block of output data for both stages. This required changes in the BOUT (Boost Output) and RVOUT (Re-entry Vehicle Output) subroutines. These changes are documented in Appendix B.

TABLE 1. COVARIANCE MATRIX OF MEASUREMENT NOISE, N

$$\begin{bmatrix} \sigma^2_{\theta} & \sigma^2_{\theta\phi} & \sigma^2_{\theta\psi} & \sigma^2_{\theta x} & \sigma^2_{\theta y} & \sigma^2_{\theta z} \\ \sigma^2_{\phi\theta} & \sigma^2_{\phi} & \sigma^2_{\phi\psi} & \sigma^2_{\phi x} & \sigma^2_{\phi y} & \sigma^2_{\phi z} \\ \sigma^2_{\psi\theta} & \sigma^2_{\psi\phi} & \sigma^2_{\psi} & \sigma^2_{\psi x} & \sigma^2_{\psi y} & \sigma^2_{\psi z} \\ \sigma^2_{x\theta} & \sigma^2_{x\phi} & \sigma^2_{x\psi} & \sigma^2_{x} & \sigma^2_{xy} & \sigma^2_{xz} \\ \sigma^2_{y\theta} & \sigma^2_{y\phi} & \sigma^2_{y\psi} & \sigma^2_{yx} & \sigma^2_{y} & \sigma^2_{yz} \\ \sigma^2_{z\theta} & \sigma^2_{z\phi} & \sigma^2_{z\psi} & \sigma^2_{zx} & \sigma^2_{zy} & \sigma^2_{z} \end{bmatrix}$$

TABLE 2. TU70 INPUTS FOR THE DATA CORRELATION FEATURE

INDICATORS:

CIN(75) - Contains an on-off "switch" for the data correlation feature (0 = Off, 1 = On)

CIN(70) - Contains an on-off "switch" for the computation of the transpose of the sensitivity matrix, S (0 = Off, 1 = On)

CONSTANTS:

CIN(499) - Contains the number of experimental data points

$$\left.\begin{array}{l} CIN(498) = \sigma_x^2 \\ CIN(497) = \sigma_y^2 \\ CIN(496) = \sigma_z^2 \\ CIN(495) = \sigma_\theta^2 \\ CIN(494) = \sigma_\phi^2 \\ CIN(493) = \sigma_\psi^2 \end{array}\right\}$$

Contains the initial inputs of the covariance matrix of the measurement noise (assumes a diagonal matrix)

$$\left.\begin{array}{l} CIN(492) = E_\theta \\ CIN(491) = E_\phi \\ CIN(490) = E_\psi \\ CIN(489) = E_x \\ CIN(488) = E_y \\ CIN(487) = E_z \end{array}\right\}$$

Contains the error-vector differences between the measured and computed values of x, y, z, θ, ϕ, and ψ

3. Instructions for Using the U70 Program

A description of the procedures for compiling, running, linking, restarting the program, and making output tapes is given in Appendix C, together with a list of the .CSS files used to accomplish these tasks, and a list of the U70 source files.

The TU70 program contains an array of input constants and indicators called CIN. Table 2 contains definitions for the constants and indicators which serve as inputs for the DATACORR subroutines for the TU70 task.

B. The TRW Program

1. Compiling the TRW Simulation Program

The TRW program was originally set up for interactive compilation, program linkage, and execution. However, in converting the program to the Perkin-Elmer 3220 computer, it became necessary to switch to batch mode compilation and linkage. The reason for turning to batch mode operation was that not every file or function required by the program is present in the program itself. Some of the necessary information is stored in independent files, and operating in batch mode permits these independent files to be found during compilation and to be linked to the MAIN program during the program linkage phase. To accomplish this result, the $BATCH command had to be placed at the beginning of the program, the $BEND command at the end of the program, and the $PROG declarations put at the beginning of each subroutine.

Next, a "compile file" was produced by modifying a copy of the systems file F7CAE.CSS (which contains the FORTRAN VII compiler and the linking loader) and then using it to compile the TRW program. Compilation time was saved by compiling one subroutine at a time, producing an object code image and saving it in a separate file, and then linking it together with other object code subroutines during the link phase. This meant that only those subroutines which had been updated had to be recompiled.

F7CAE.CSS is a procedure which compiles and links any program. The compiler in F7CAE.CSS uses system file F7D.TSK to produce an object file and F7D.ERR to record errors found during the compile phase. The F7CAE.CSS link loader processes object files generated by the compiler and creates a task file for execution. The link loader also creates a load map and provides a record of any undefined external references or symbols. Appendix D lists the file names used by the TRW program and Appendix E lists the program's subroutine names and calling sequences.

Of the many error messages which surfaced during the first compilation, a substantial fraction was associated with VAX-peculiar FORTRAN INCLUDE statements. INCLUDE is a VAX FORTRAN compile-time command which causes the compiler to copy a disk file into a program - typically, a file containing COMMON and EQUIVALENCE statements. This permits a simple INCLUDE statement to be substituted in subroutines for the extensive COMMON and EQUIVALENCE statements that would ordinarily be found there instead. This practice not only reduces the amount of effort required to write and update the program but also reduces the chances of making a mistake in coding, since the COMMON and EQUIVALENCE statements only have to be updated at one place in the program rather than in every subroutine in which they are referenced.

Another VAX-peculiar FORTRAN VII enhancement is the DO WHILE command. Since the DO WHILE statement is not recognized by Perkin-Elmer's FORTRAN VII compiler, it was necessary to replace DO WHILE's in the original program with ordinary DO statements in the Perkin-Elmer version of the program.

The above modifications constitute the principal programming changes that have so far had to be made to the TRW subroutines in order to convert them from the VAX 11/780 computer to the Perkin-Elmer 3220 computer.

2. Subroutines Missing From the TRW Program

As mentioned in Section II.E., the TRW simulation program was delivered with some major portions of the program missing. The most critical missing subroutine was INTERP(Interpolation) which, by interpolating large data tables, would have provided the thrust characteristics, mass properties, and aerodynamic data needed to simulate the flight. The main control program, MAIN, was also missing, although a program SDCTRL (Six-Degree-of-Freedom Control) was found which seems to perform similar functions. Essential portions of the program that were required to simulate steering, guidance, and navigation in the Pershing Airborne Computer were also missing. Finally, an error message routine ERRMSG was missing.

a. INTERP(Interpolation)

With the aid of a former developer of the TRW INTERP routine, a prior version of INTERP was located in an earlier edition of the TRW simulator and efforts were made to understand it and use it in the current version of the TRW program. At the heart of the INTERP routine is the above mentioned involved set of aerodynamic tables, organized in a way that minimizes storage requirements without unduly increasing run times. A discussion of what has been learned about this multi-dimensional data table is presented in Appendix F.

b. MAIN and SDCTRL (Six-Degree-of-Freedom-Control) Programs

After finding that the MAIN routine was missing from the TRW simulator, efforts were begun to develop such a function-controlling routine. An earlier version of MAIN was found in a program listing and the listing was used to help understand what was needed to recreate a current MAIN routine. A flowchart of this early version MAIN routine is contained in Appendix G. Appendix H contains a flowchart of the SDCTRL routine.

c. The Output Processor

The TRW simulation program has an output processor which takes advantage of large arrays to store values of variables for output. These arrays are saved in three groups named BIGØ1R, BIGØ1I, and BIGØ1D, which contain Real, Integer, and Double Precision variables, respectively. These output-variable values are transferred to the BIGØ1 arrays for use by the output processor through EQUIVALENCE statments located in each subroutine [1]. Each variable stored in BIGØ1R, BIGØ1I, and BIGØ1D is stored on a disk file BIGØ1.DAT. The names of these variables are listed in Appendix I.

3. Tape-to-Disk Conversion of the TRW Program

TRW delivered its simulation program to the System Evaluation Branch on a nine-track 1600 bit-per-inch tape. This was loaded onto a Perkin-Elmer 3220 disk pack using the following procedure:

Mount the tape on a 1600 bpi tape drive.

```
> COPY
> AL FILE1.FTN,IN,80
> OUT FILE1.FTN
> COPY *,*
```

IV. RESULTS

A. Flight Safety (Motor Nozzle Deflection) Results

As mentioned earlier in Section II,B, this Pershing II missile simulation was carried out using the U70 simulator to determine what would happen if, through some hardware failure, the nozzle were accidently deflected during a live missile firing. In carrying out these simulated flight tests, the missile was allowed to "fly" unperturbed (in the computer) for a short time. Then the motor nozzle was deflected by a pre-determined angle and the simulation continued until either the missile's total angle of attack exceeded 15° or the normal (perpendicular to the body of the missile) acceleration exceeded 5 g's. Either one of these occurences was assumed to generate sufficiently unstable conditions that breakup of the missile would occur, so the simulation was terminated at that point.

The spurious nozzle deflections were assumed to occur at 30 seconds and at 49 seconds into the flight. The 30-second flight time was chosen because, at 30 seconds, the aerodynamic forces acting on the missile would be at or near their maximum, i.e, a worst-case condition. The 49-second flight time was chosen because it is almost at first-stage burnout, and is a time when missile failure is likely. Three nozzle deflection angles were tried, 0.5°, 2°, and 7.6°, first in pitch and then in yaw, leading to a total of twelve cases (two deflection times and three nozzle deflection angles, first in pitch and then in yaw). All the cases were simulated flights from McGregor, New Mexico, to McDonald's Well, New Mexico, with the targeting conditions given in Table 3 below. Before presenting these results in detail, some data comparison problems need to be discussed.

TABLE 3. LAUNCH SITE AND TARGET

McGregor, NM	LAUNCH SITE
	Latitude = 32.09575°
	Longitude = -106.2035°
	Altitude = 1251.0000 m
McDonald's Well, NM	TARGET SITE
	Latitude = 33.113580°
	Longitude = -106.35897°
	Altitude = 1228.00000 m

In order to provide a standard by which to compare the deviated-nozzle runs, a nominal or baseline simulation was run in which it was assumed that the missile followed a nominal, undeflected flight plan. Since the deviated nozzle results were printed at 0.1 second intervals, it would have been advantageous to have also printed the baseline results at 0.1 second intervals for purposes of comparison. Unfortunately, the large volume of printout generated by the U70 program made it impractical to print results at 0.1 second intervals, and the results were available only at 1.0 second intervals. Consequently, the baseline results had to be interpolated at 0.1 second intervals in order to compare them to the deviated-nozzle results, i.e., a direct comparison was not possible.

A second data comparison problem resides in the fact that, for unknown reasons, the first data points in the deviated-nozzle simulation printouts (the results for 30.0 seconds and the results for 49.0 seconds) are invalid. These 30- and 49-second numbers are not random but seem to come from some earlier time in the flight. Subsequent results appear to be correct when compared to the unperturbed flight results. For example, the interpolated unperturbed flight results for 30.1 seconds agree to four decimal places with the deviated-nozzle results, with subsequent data departing from the baseline data in a regular and reasonable way. It is the authors' conclusion that the faulty first point in each set of numbers was the result of a print-routine malfunction rather that an error in the deviated nozzle results. However, the reader needs to be aware of this characteristic in the data.

Figure 1 depicts the missile's flight path for an unperturbed flight plan and for the three different yaw-axis nozzle deflection angles looking down from above when the nozzle deflections occur 30.0 seconds into the flight. Figure 1 is a plot of downrange coordinates versus crossrange coordinates. The "straight line" in Figure 1 represents the path of an unperturbed missile. (It curves about 1° to the right but the curvature is too small to be visible in the plot.) For all three deflections, the missile disintegrates because the yaw angle of attack exceeds 15° rather than because the normal acceleration exceeds 5 g's. Note that for the smaller the nozzle deflection angle, the farther the missile can travel before it reaches the critical angle of 15°.

Figure 2 shows a similar plot when the nozzle deflection occurs 49 seconds into the flight. Note that the flight path deflections appear to be smaller than they were when the nozzle deflection occurred at 30 seconds (Figure 2). In reality, the deflections are the same but the velocity of the missile is greater, and this increases the horizontal scale of the plot. (The results for a nozzle deflection of 7.6° are missing because of a faulty computer run.)

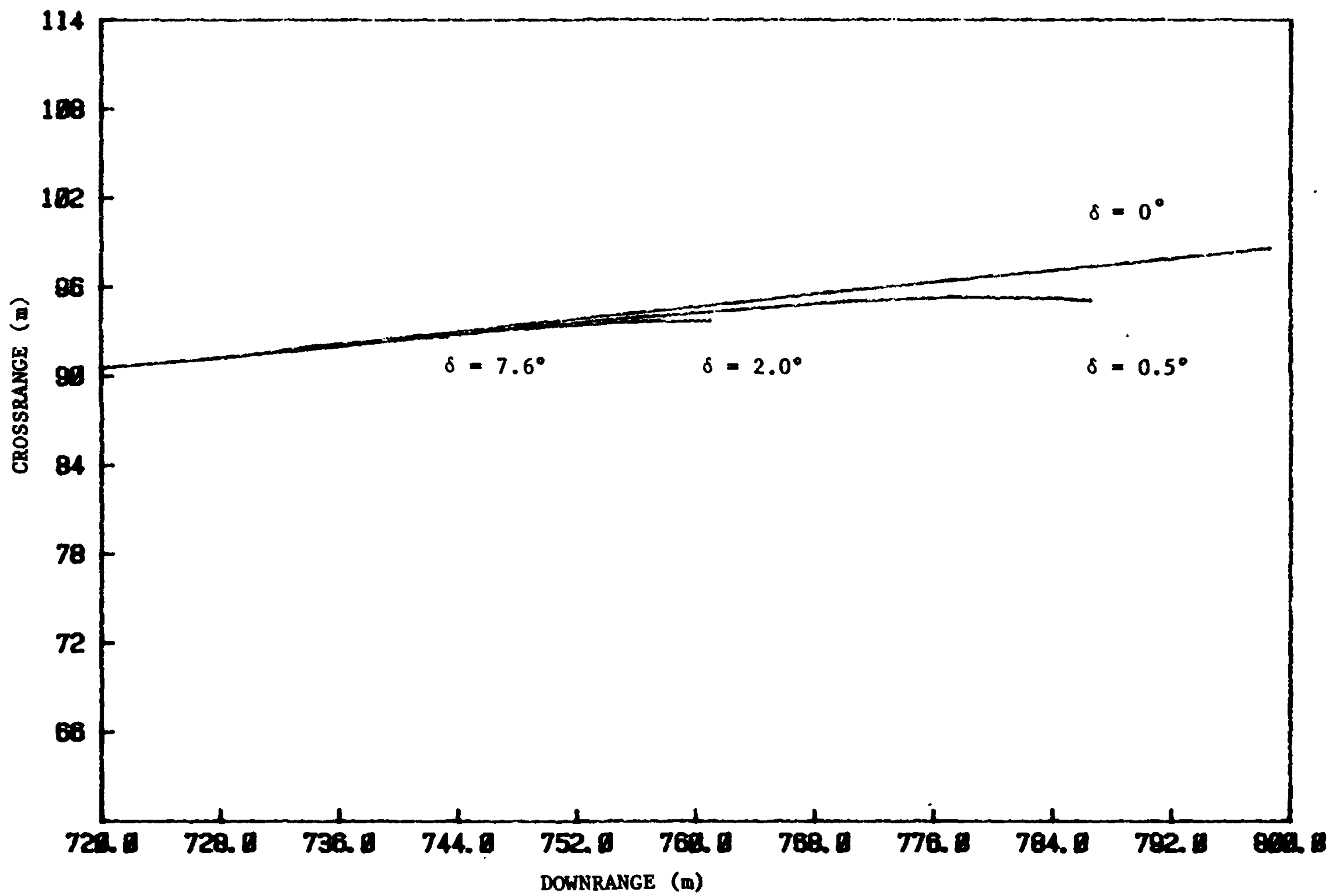

Figure 1. Crossrange vs. Downrange for Yaw Nozzle Deflections (δ's) Occurring at 30 Seconds into the Flight.

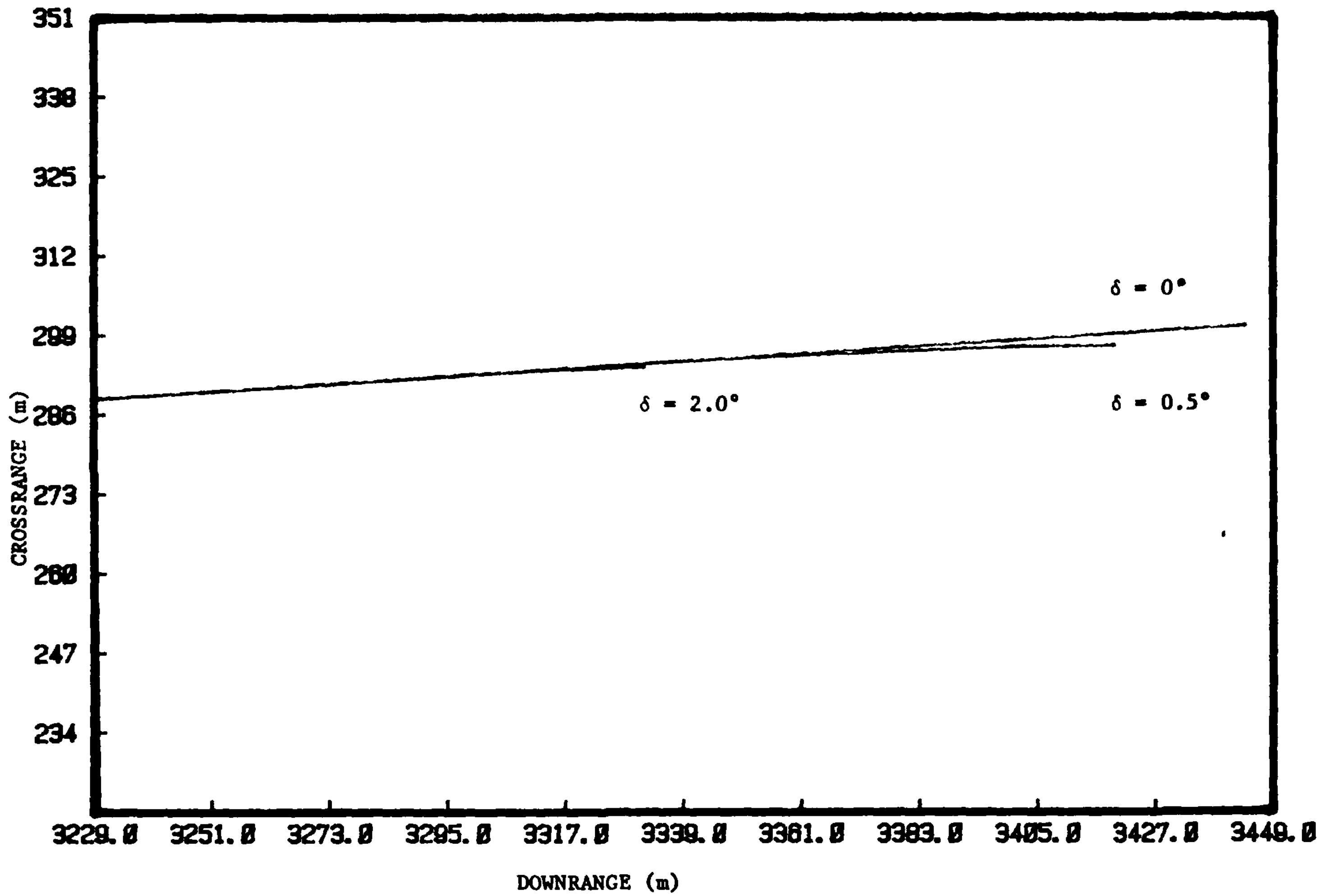

Figure 2. Crossrange vs. Downrange for Yaw Nozzle Deflections (δ's) Occurring at 49 Seconds into the Flight.

Figure 3 shows a plot of the missile vertical pitch angle versus its downrange coordinates when a nozzle deflection in pitch occurs at 30 seconds, and Figure 4 depicts similar plots for the 49 second pitch nozzle deflection.

Figures 5 through 8 show how the normal (lateral) accelerations of the missile vary with time. An examination of these curves confirms that in every case, the simulated flights terminated before the normal accelerations exceeded the destructive limit of 5 g's, i.e., the flights terminated because the missile's total angle of attack exceeded 15°. There is an interesting tendency for these normal accelerations to rise, dip, and then rise again. No simple explanation has been advanced for this behavior, although the Pershing II guidance system is sufficiently complex that a complex response might be expected in the event of a major nozzle deflection malfunction. (The results for a nozzle deflection of 7.6° are missing because of the faulty computer run mentioned previously in connection with Figure 2.)

These simulations show that the missile would leave its allowable flight corridor and exceed the allowable angle of attack within approximately one second.

B. Aerodynamic Model Validation Results

As mentioned in Section II.B., this study could not be completed because Pershing II flight test data were not available in time.

In support of this task, the DATACORR subroutine, incorporating the Maximum Likelihood Method of data correlation, was written, compiled, and, insofar as possible, was tested. However, one important part of this subroutine, the derivation of sensitivity coefficients, was not completed.

C. Tactical Ballistic Missile Trajectory Results

As mentioned in Section II.C., nine U70 runs were made for this study for nine input trajectory profiles. Three of these profiles entailed offsets in target latitude and longitude.

When the study was completed, the results were output on tape and delivered to the sponsor.

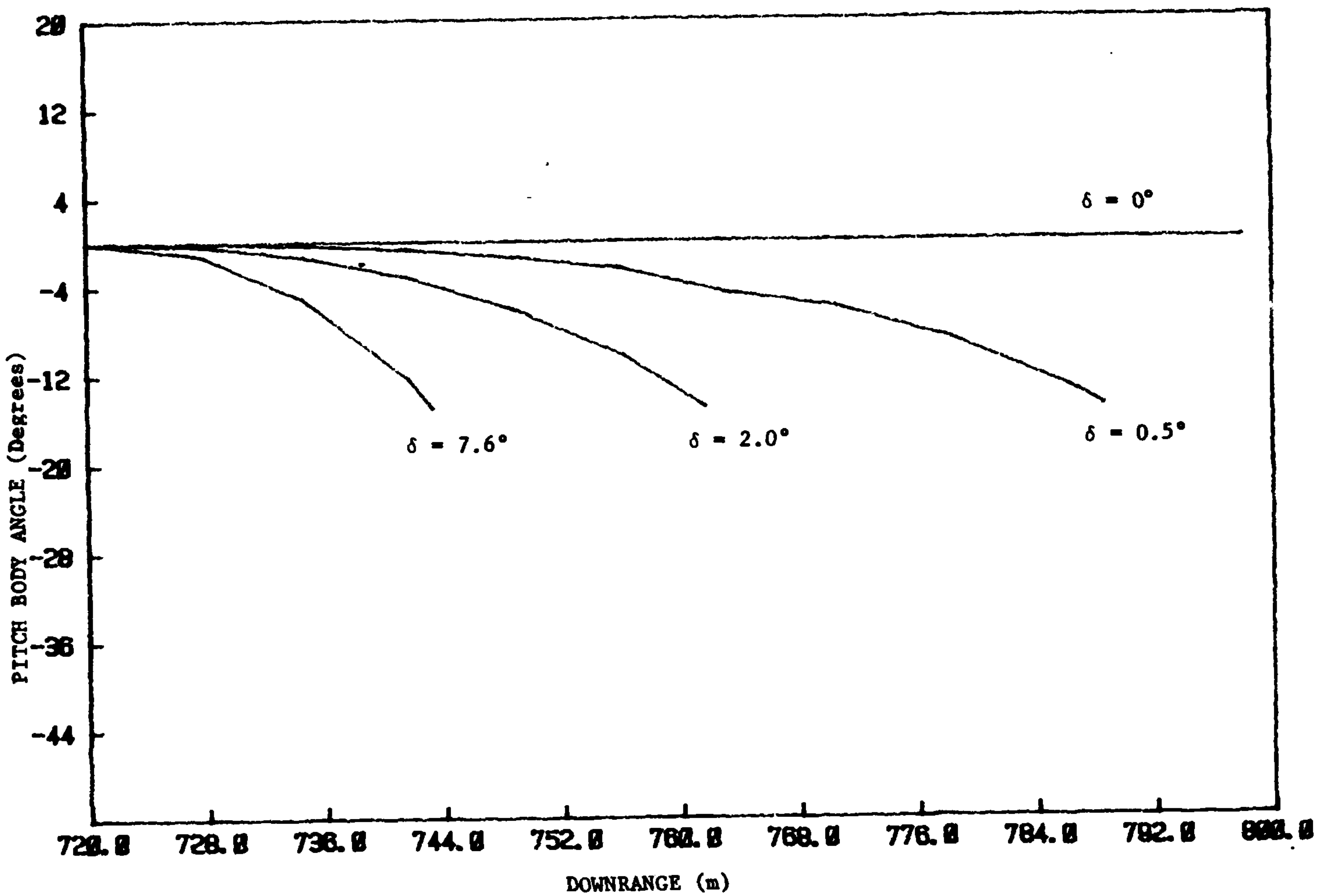

Figure 3. Pitch Body Angle vs. Downrange for Pitch Nozzle Deflections (δ's) Occurring at 30 Seconds into the Flight.

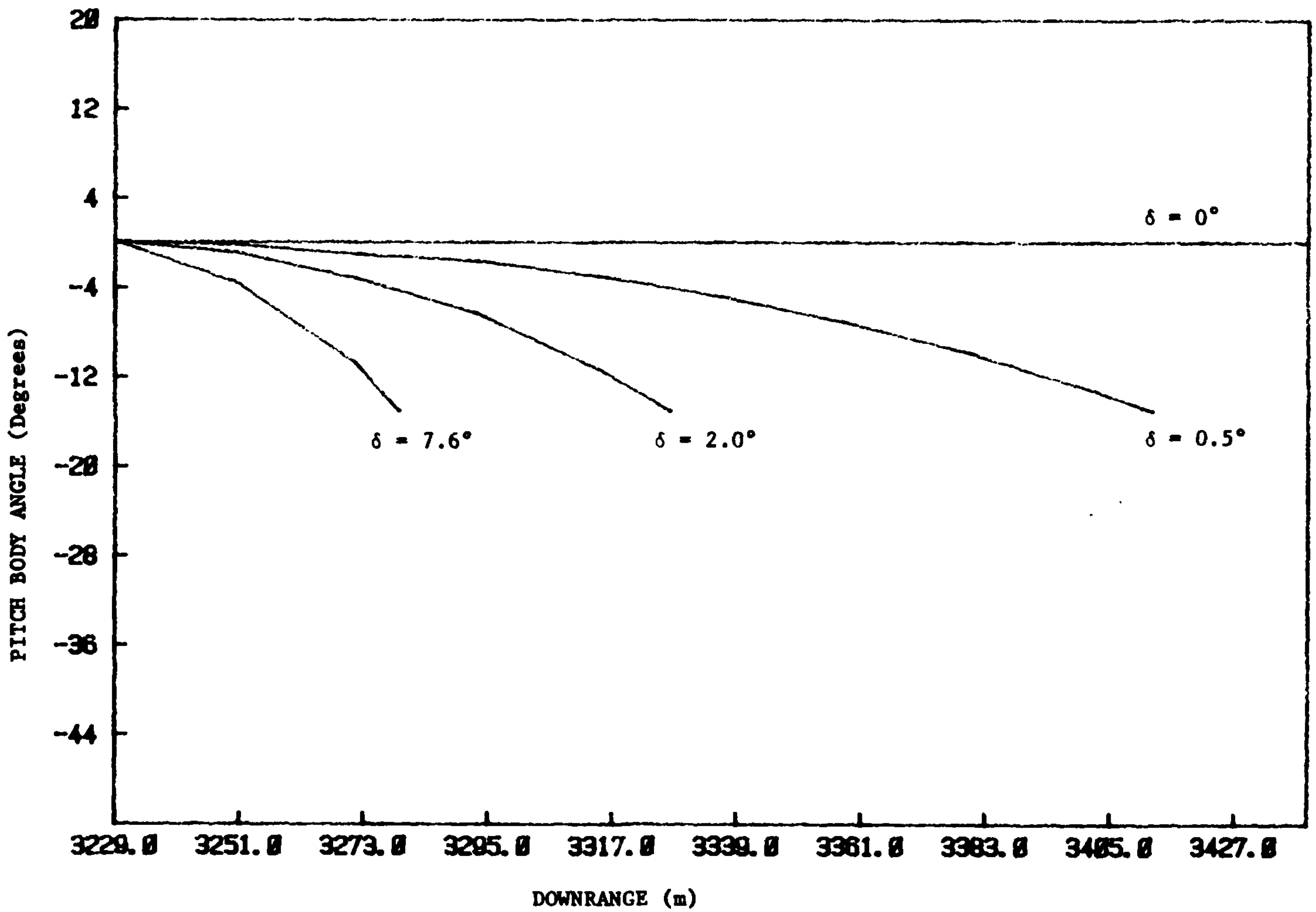

Figure 4. Pitch Body Angle vs. Downrange for Pitch Nozzle Deflections (δ's) Occurring at 49 Seconds into the Flight.

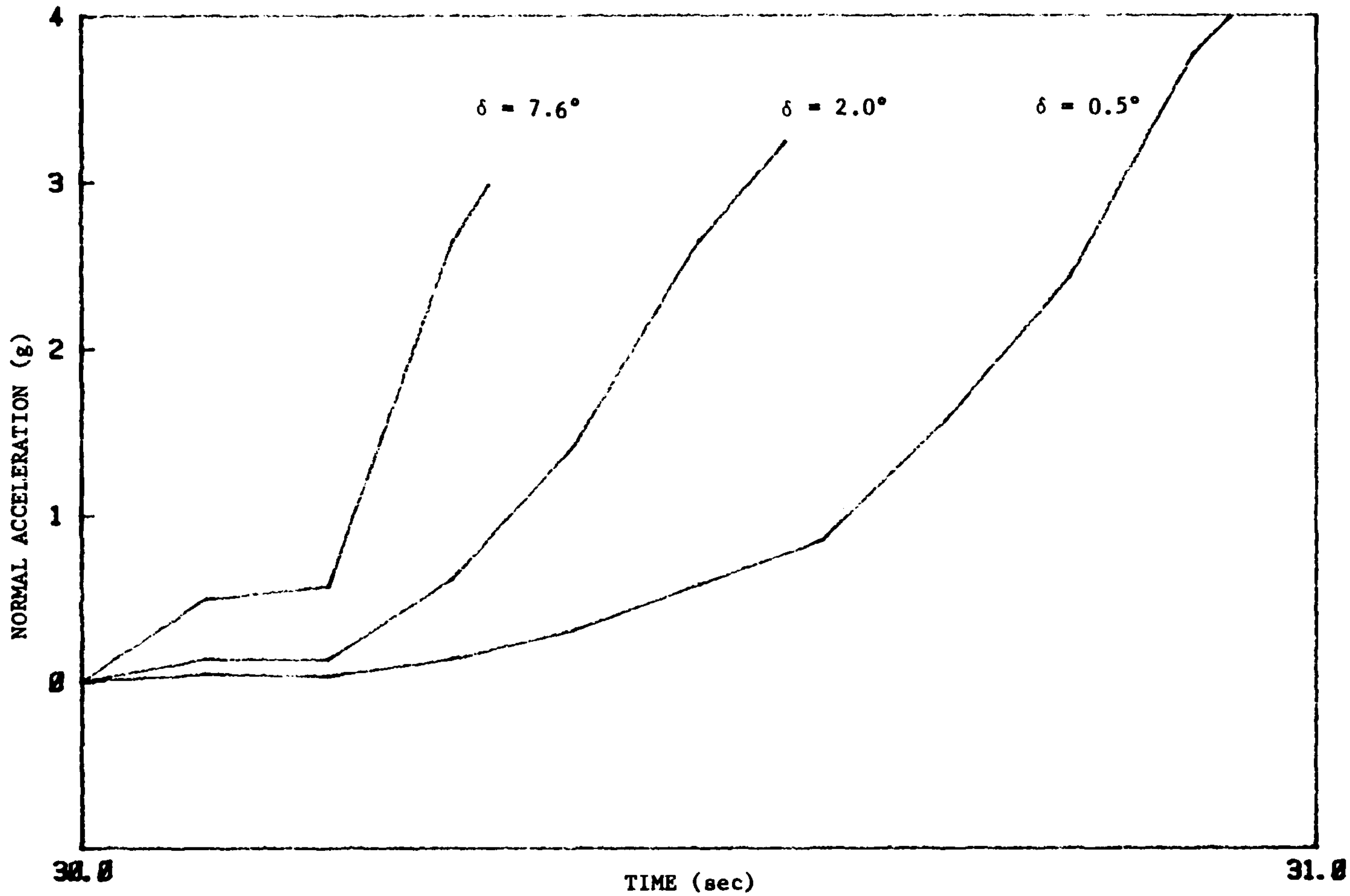

Figure 5. Normal Accelerations vs. Time for Pitch Nozzle Deflections (δ's) Occurring at 30 Seconds into the Flight.

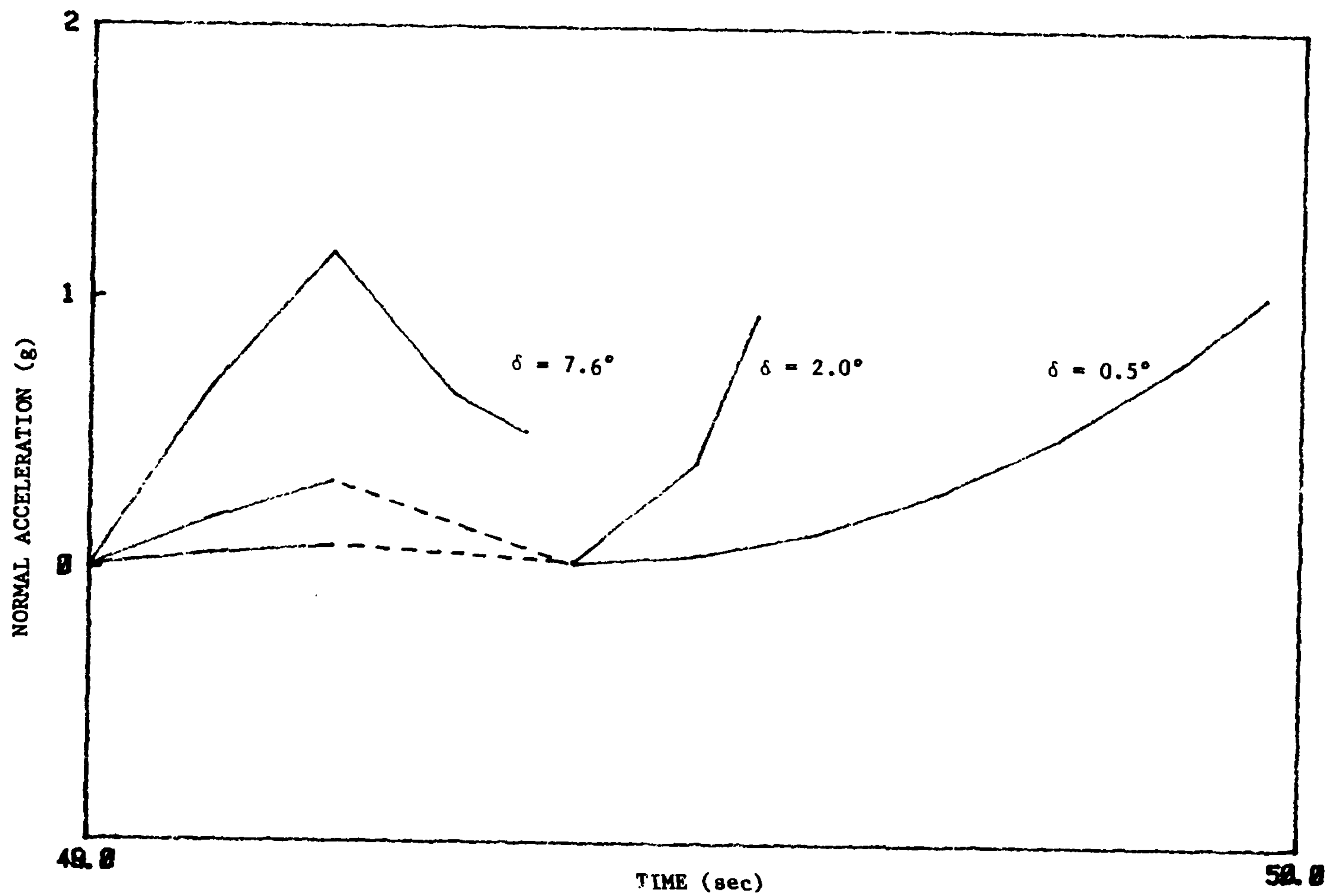

Figure 6. Normal Accelerations vs. Time for Pitch Nozzle Deflections (δ's) Occurring at 49 Seconds into the Flight.

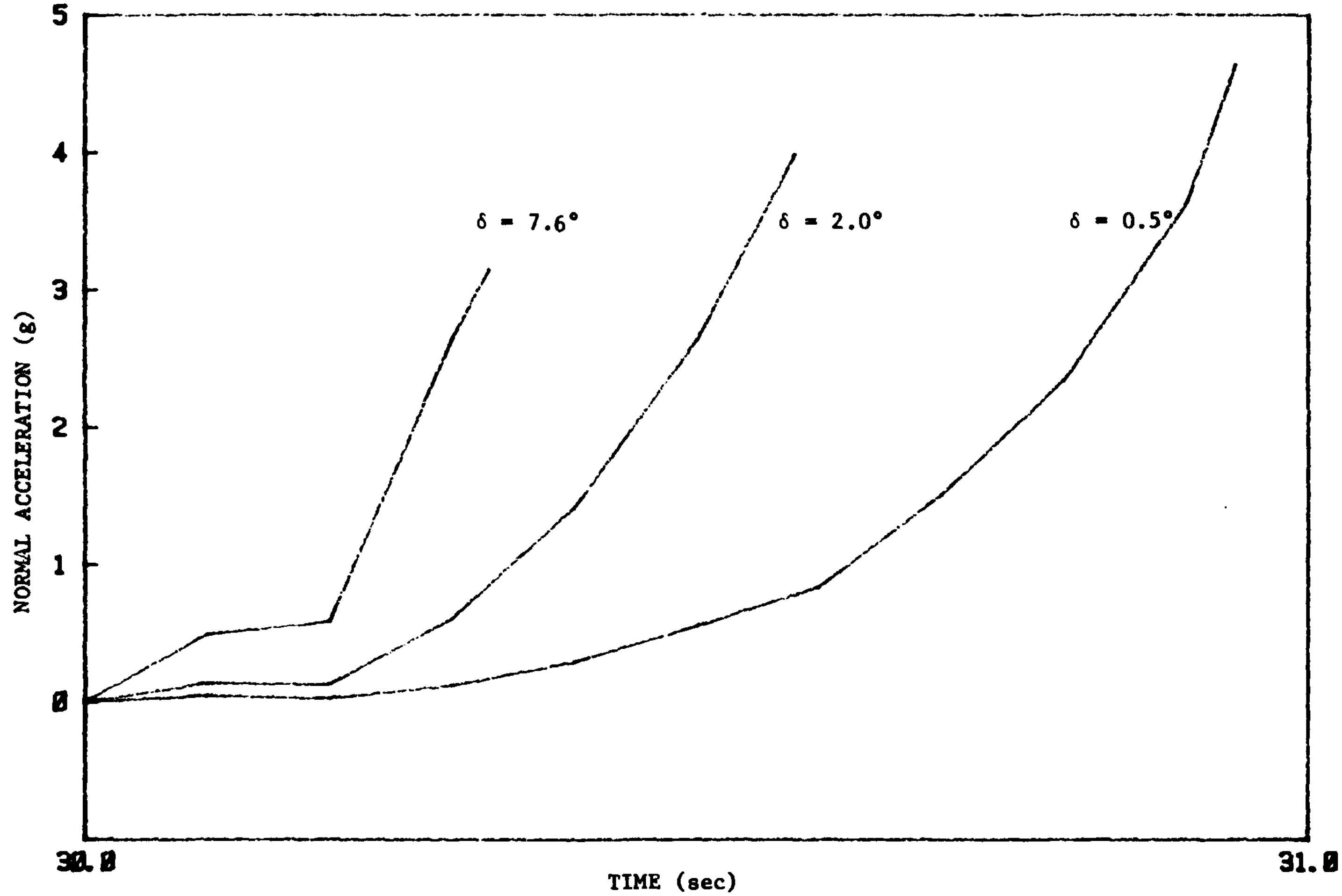

Figure 7. Normal Accelerations vs. Time for Yaw Nozzle Deflections (δ's) Occurring at 30 Seconds into the Flight.

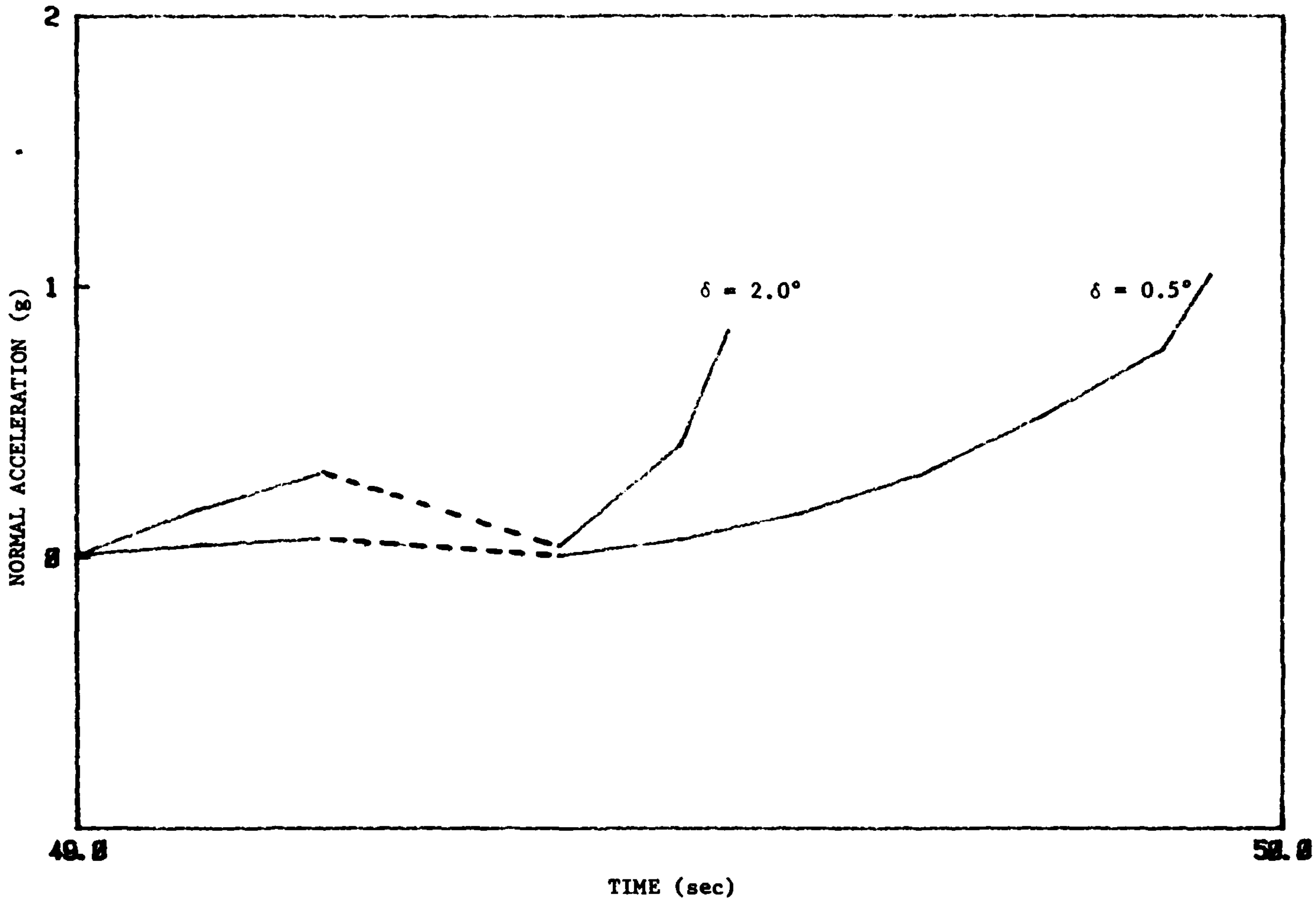

Figure 8. Normal Accelerations vs. Time for Yaw Nozzle Deflections (δ's) Occurring at 49 Seconds into the Flight.

D. Conversion and Installation of the TRW Simulation Program

Some of the details of the TRW program conversion have been discussed in Section III.B., and in Appendices D through I. A summary of the results is as follows:

1. All subroutines except those missing from the incoming program tape were converted and successfully compiled.

2. The organization of the INTERP tables was deciphered and documented (Appendix F).

3. The MAIN and SDCTRL subroutines were flowcharted (Appendices G and H).

4. A linking task builder TRWLINK.CSS and a compiler task builder DAVEF7.CSS were created to accommodate the TRW program.

V. RECOMMENDATIONS FOR FURTHER WORK

A. Flight Safety (Motor Nozzle Deflection) Recommendations

One of the recommendations for future work is to carry out these evaluations, re-running the simulation with print statements that monitor the moment-by-moment variations in the angles of attack, the down-range and cross-range coordinates, and the inertial flight path angles before, during, and after the moment when the nozzle deviation occurs.

B. Aerodynamic Model Validation Recommendations

Recommendations for future work consist of completing the DATACORR subroutine and carrying out the study using actual flight data.

Normally, in employing the Maximum Likelihood Method, all the computed constitutive forces and moments that make up the total X, Y, or Z forces or ϕ, θ, or ψ moments (constituents of lift, drag, etc.,) are compared individually, with their equivalent measured values. However, the aerodynamic model used in the U70 simulator is not yet sufficiently detailed to permit making these comparisons because of the difficulty of attributing corrections to individual components of the U70 predictions, so the comparisons are to be made at the level of overall body forces and torques. For this reason, the DATACORR subroutine doesn't presently output its force and torque corrections to the main U70 program. However, future plans call for the upgrading of the main program to provide aerodynamic modeling at the constitutive level. When that is done, the DATACORR subroutine and the TU70 task will be upgraded to output these corrections to the U70 program.

C. Tactical Ballistic Missile Trajectory Recommendations

Recommendations for further work depend upon the sponsor's requirements.

D. Conversion and Installation of the TRW Simulation Program

The Maximum Likelihood Method correlation technique being used to validate the U70 simulator predictions against actual flight data (Section II.C.) should be used to evaluate the TRW simulator.

Recommendations for future work consist of completing the conversion and installation of the TRW program, using it to simulate Pershing II test flights, and comparing its results with the actual test flight data.

REFERENCES

[1] "U70 Utilization Report," Martin-Marietta Corporation, Orlando, Florida, Revision A, January 27, 1981.

[2] "Computer Program Missile Interim Detail Specification/Performance/ Design Requirements for Pershing Airborne Computer Program for Pershing II Weapon System, Engineering Development Program," Specification MIS-21748, Martin-Marietta Corporation, October 1, 1980.

APPENDIX A
CODE LISTING FOR THE DATACORR (Data Correlation) SUBROUTINE

TABLE A-1. CODE LISTING FOR THE DATACORR SUBROUTINE

```
$PROG DATACORR
      SUBROUTINE DATACORR
$INCLUDE 9,U70TIM.COM
C ***     BEGIN COMMON PACK    ***
 IMPLICIT DOUBLE PRECISION (A-H,O-Z)
 REAL C3V,CUR3V,DCB,DCI,SCD,SCI,PSAVE
 COMMON CIN(400),SCD(1100),SCI(1100),DCB(7200),DCI(640)
 COMMON TITL(12),TMAJ(30),TAB(50),TTAB(90),CUR3V(3300),DUM2(10)
 COMMON ICON(100),IDB(32),IDC(33),IDI(32),ISC(31),ISI(30),IND3V(1
 COMMON NDB(32),NDI(32),NSI(30),NSTR(30),NTV(3,10),IALIGN
 COMMON /CONST/ A13    ,A23    ,A33    ,COSBL  ,COSPL  ,CPSIA  ,CTHA   ,
1                CXNUP  ,CXNUY  ,DTCR   ,D2AC2  ,F11    ,F12    ,F13    ,
2                F21    ,F22    ,F23    ,F31    ,F32    ,F33    ,RL     ,
3                RPRES  ,RTOD   ,SINBL  ,SINPL  ,SPSIA  ,STHA   ,SXNUP  ,
4                SXNUY  ,S11    ,S12    ,S13    ,S21    ,S22    ,S23    ,
5                S31    ,S32    ,S33    ,TANPL  ,TOLH   ,TOL1D  ,TOL2D  ,
6                U11    ,U21    ,U22    ,U23    ,U31    ,U33    ,PI     ,
7                COT    ,SOT    ,PIO2   ,TWOPI  ,THIRD  ,SQRT2  ,SQRT3  ,
8                PIO4

 COMMON ACA    ,ACCEL  ,ACR    ,ACR4   ,ACR5   ,BPINT  ,BYINT  ,CGDE
 COMMON DGAMS  ,DGGT   ,DRHO   ,DVEL   ,GAMMC  ,GAMME  ,GAMRE  ,IGUID
 COMMON IPCD   ,ITHM   ,IYAW   ,PHIPE  ,PHIRE  ,PHIYE  ,PHPEC  ,PHPEP
 COMMON PHYEC  ,PHYEP  ,PSIC   ,PX     ,PY     ,PZ     ,RCX    ,RCY
 COMMON RCZ    ,RD     ,RDD    ,RMAG   ,SGDE   ,SGE    ,SGG    ,TFF
 COMMON TFFS   ,TGE    ,TGN    ,TGS    ,THETAC ,TIMPL  ,URANX  ,URANY
 COMMON URANZ  ,URAX   ,URAY   ,URAZ   ,URAZX  ,URAZY  ,URAZZ  ,URX
 COMMON URY    ,URZ    ,VE     ,VEX    ,VEXS   ,VEY    ,VEYS   ,VEZ
 COMMON VEZS   ,VIP    ,VLEX   ,VLEY   ,VLEZ   ,VR     ,VRE    ,VREH
 COMMON VREHX  ,VREHY  ,VREHZ  ,ZDIFF  ,ZETAE  ,ZETRE  ,ZINTC  ,ZINTP
 COMMON ZINTS

 COMMON AA1    ,AA2    ,ALDDTC,ALDDTS,ALOADF,ALOADS,ALPDDE,ALPHDD
 COMMON ALPHP  ,ALPHPD,ALPHPS,ALPHS  ,ALPHT  ,ALPHY  ,ALPHYD,ALPHYS
 COMMON ATC    ,ATP    ,AX     ,AY     ,AZ     ,A11    ,A12    ,A21
 COMMON A22    ,A31    ,A32    ,BB2    ,BEDDTC,BEDDTS,BETADD,BETAP
 COMMON BETAR  ,BETAY  ,BETAW  ,BETDDE,BTC    ,BTP    ,B11    ,B12
 COMMON B13    ,B21    ,B22    ,B23    ,B31    ,B32    ,B33    ,CA
 COMMON CAAP   ,CAAY   ,CAIN   ,CLP    ,CMQ    ,CMXP   ,CN     ,CNAPD
 COMMON CNAYD  ,CNP    ,CNG    ,CNVDO  ,CNVDP  ,CNY    ,CNZ    ,COSPSD
 COMMON COSTHD,CPHI   ,CPSI   ,CTHE   ,CY     ,CYP    ,CYR    ,C11
 COMMON C12    ,C13    ,DELTA1,DELTA2,DELTA3,DELTA4,DFP    ,DFR
 COMMON DFY    ,DT     ,DVTC   ,DVTP   ,DVTS   ,D11    ,D12    ,D13
 COMMON D21    ,D22    ,D23    ,D31    ,D32    ,D33    ,DELC   ,EDLC
 COMMON EDDLC  ,EPHDC  ,EPHDS  ,EPHIC  ,EPHIP  ,EPHIS  ,EPSDC  ,EPSDS
 COMMON EPSIC  ,EPSIP  ,EPSIS  ,ETA    ,ETHAC  ,ETHAP  ,ETHAS  ,ETHDC
 COMMON ETHDS  ,FE     ,FP     ,FROLL  ,FSL    ,FT     ,FX     ,FXM
 COMMON FY     ,FYAV   ,FYAW   ,FYM    ,FZ     ,FZAV   ,FZM    ,G
 COMMON GA     ,GADDTC,GADDTS,GAMDD  ,GAMDDE,GR     ,GTC    ,GTP
 COMMON H      ,HDT    ,H11    ,H12    ,H13    ,H21    ,H22    ,H23
 COMMON H31    ,H32    ,H33    ,IAERO  ,ICN    ,ICO    ,IDINT  ,IDISC
```

TABLE A-1. CODE LISTING FOR THE DATACORR SUBROUTINE (Continued)

```
COMMON IDT   ,IDUBL ,IENV  ,IFAIL ,IFCON ,IFINC ,IFLAG ,IGC
COMMON IHIJ  ,IHT   ,IMAJ  ,IMISS ,INDH  ,INDT  ,INT   ,IPAGE
COMMON IPLOT ,IPNT  ,IPRNT ,IPROP ,IRJU  ,IRUN  ,ISEN  ,ISMTH
COMMON ITA   ,ITB   ,ITC   ,ITHR  ,ITOLB ,ITPOS ,ITT   ,IUNT
COMMON IUPD  ,IVAC  ,IWD   ,IWIN  ,JSTG  ,KFLAG ,KSTG  ,LBCNT
COMMON LINC  ,LINTOT,MAJ   ,MBAL  ,NTPU  ,P     ,PC    ,PDC
COMMON PDS   ,PDT   ,PHI   ,PHID  ,PHIE  ,PHILC ,PHIMC ,PP
COMMON PS    ,PSI   ,PSID  ,PSIE  ,PTIM  ,PXPC  ,PXPP  ,PXPS
COMMON PYPC  ,PYPP  ,PYPS  ,PZPC  ,PZPP  ,PZPS  ,P11   ,P12
COMMON P13   ,P21   ,P22   ,P23   ,P31   ,P32   ,P33   ,Q
COMMON QA    ,QC    ,QDC   ,QDS   ,QMIN  ,QP    ,QS    ,R
COMMON RAIM  ,RC    ,RDC   ,RDS   ,RET   ,RHO   ,RP    ,RS
COMMON SIGMAP,SIGMAY,SINPSD,SINTHD,SPHI  ,SPSI  ,STHE  ,SXGZGD
COMMON TBO   ,TBS   ,TDIFF ,TDT   ,TEMP1 ,TEMP2 ,TEMP3 ,THETA
COMMON THETAD,THETAE,THETL ,TIMC  ,TIMP  ,TIMS  ,TLIM  ,TST
COMMON TSTMAX,TSTMIN,TTT   ,UMA   ,VIXC  ,VIXP  ,VIXS  ,VIYC
COMMON VIYP  ,VIYS  ,VIZC  ,VIZP  ,VIZS  ,VMA   ,VN    ,VPRIX
COMMON VPRIY ,VPRIZ ,VRW   ,VRXP  ,VRYP  ,VRZP  ,VS    ,VW
COMMON VWE   ,VWS   ,VX    ,VY    ,VZ    ,W     ,WACB  ,WACBS
COMMON WD    ,WDACB ,WDACBS,WDS   ,WG    ,WIN   ,WMA   ,WPB
COMMON WPBS  ,WPN   ,WPS   ,WS    ,XC    ,XDC   ,XDDC  ,XDDS
COMMON XDG   ,XDIN  ,XDP   ,XDS   ,XEL   ,XET   ,XIN   ,XIXX
COMMON XIYY  ,XIZZ  ,XKN   ,XLAM  ,XLAMM ,XLCM  ,XLCP  ,XL11
COMMON XL12  ,XL13  ,XL21  ,XL22  ,XL23  ,XL31  ,XL32  ,XL33
COMMON XM    ,XMAID ,XMASS ,XMAX  ,XMAY  ,XMAZ  ,XMAVX ,XMAVY
COMMON XMAVZ ,XMC   ,XMFX  ,XMFY  ,XMFZ  ,XMSP  ,XMTX  ,XMTY
COMMON XMTZ  ,XMXACS,XMYACS,XMZACS,XNLOAF,XNU   ,XNV1  ,XNV2
COMMON XNV3  ,XP    ,XPC   ,XPDC  ,XPDD  ,XPDS  ,XPP   ,XPS
COMMON XS    ,YC    ,YDC   ,YDDC  ,YDDS  ,YDG   ,YDIN  ,YDP
COMMON YDS   ,YEL   ,YET   ,YMAID ,YP    ,YPC   ,YPDC  ,YPDD
COMMON YPDS  ,YPP   ,YPS   ,YS    ,ZC    ,ZDC   ,ZDDC  ,ZDDS
COMMON ZDG   ,ZDIN  ,ZDP   ,ZDS   ,ZEL   ,ZET   ,ZIN   ,ZMAID
COMMON ZP    ,ZPC   ,ZPDC  ,ZPDD  ,ZPDS  ,ZPP   ,ZPS   ,ZS
COMMON ZZ1   ,ZZ2   ,ZZ3   ,ZZ4   ,ZZ5   ,ZZ6   ,ZZ7   ,ZZ8

COMMON ACC   ,ADE11 ,ADE12 ,ADE13 ,ADE21 ,ADE22 ,ADE23 ,ADE31
COMMON ADE32 ,ADE33 ,ALDC  ,ALDCL ,ALPHPL,ALPHYL,ALPC  ,ALRC
COMMON ALYC  ,ALRFP ,ALRFY ,ALTC  ,ALTCL ,AMAX  ,APEST ,AYEST
COMMON APX   ,APY   ,APZ   ,BERR  ,BETA  ,BETASH,BETASW,CAO
COMMON CABASE,CASF  ,CNLIM ,CUN   ,CUE   ,CUD   ,DELPA ,DELRA
COMMON DELYA ,DELPL ,DELRL ,DELYL ,DLAM  ,DLAMD ,DVN   ,DVE
COMMON DVD   ,ENSCS ,EESCS ,ERALT ,GADPC ,GADDPC,GADDPS,GDDDP
COMMON GADYC ,GADDYC,GADDYS,GDDDY ,GAMMAD,GAMMAT,GDDC  ,GDDCL
COMMON GDPC  ,GDRC  ,GDYC  ,GDPX  ,GDPY  ,GDPZ  ,GDTC  ,GDTCL
COMMON GLIM  ,HABS  ,HDOT  ,HNAV  ,HPOS  ,IASW  ,IATM  ,IBSW
COMMON IDUM  ,IFIL  ,IGAIN ,ILAG  ,ILAST ,IRREF ,IRUP  ,ISCS
COMMON ISEQ  ,ISHD  ,ISTER ,JBSW  ,NREP  ,PHIAL ,PHICMD,PHIGD
COMMON PHITAB,PN    ,PE    ,PD    ,POS   ,PUN   ,PUE   ,PUD
COMMON RKE   ,RKGD  ,RKGT  ,RUX   ,RUY   ,RUZ   ,TEJ   ,TG
COMMON TLAM  ,TLAMD ,TPD   ,TRTN  ,UPC   ,URC   ,UYC   ,VHS
COMMON VNAV  ,VXS   ,VYS   ,VZS   ,WAP   ,XFAIL ,YFAIL ,ZFAIL
```

```
 COMMON XMT    ,XMTD  ,YMT    ,YMTD  ,ZMT    ,ZMTD

     END COMMON PACK   ***
 COMMON/OTHERS/AC      ,ACM    ,ACPX   ,ACPY   ,ACPZ   ,ACX    ,ACY
1ACZ    ,API(2,14)     ,APC    ,APFB   ,AYC    ,AYFB   ,CGE    ,CLDR
2COSTC  ,DGAMC  ,DNV(2,6)      ,EPP    ,EYP    ,FEEP   ,GAMMCS,GAMMG
3IACFLG,IAPFLG,IBO     ,IDB3   ,IDB4   ,IDB5   ,IDB6   ,IDI3   ,IDI4
4IDI5   ,IDI6   ,ICOF   ,IGAFLG,IPOFL  ,PMX    ,PMY    ,PMZ    ,QAP
5SINGC  ,SINPC  ,SINTC  ,TGSM   ,TGSMS  ,TTIMP  ,TVACF  ,TVACP  ,T11
6T12    ,T13    ,T21    ,T22    ,T23    ,T31    ,T32    ,T33    ,VG
7VIS    ,VTS    ,XMAP   ,XMSAP  ,YIN    ,HANK(20)       ,ACSX
 DIMENSICN DMG(64),PMG(64)
 DIMENSION DCG(54),PCG(54)
 DIMENSICN AS(6,6),SAS(6,6),AA(6,6),AAT(6,6),AS1(6,6)
 DIMENSICN AL(6,6),AM(6,6),AM1(6,6),AE(6,1),AN(5,1)
 DIMENSICN DC(6,1),AP(6,1),AR(6,1),AAR(6,1)
 DIMENSICN X(6,12),WORK(12),IHLD(6)
 DATA DMG/5H   THE,5H   PHE,5H   PSE,5H    XE,5H    YE,5H    ZE,
1         5H   THC,5H   PHC,5H   PSC,5H   XCC,5H   YCC,5H   ZCC,
2         5H  THDC,5H  PHDC,5H  PSDC,5H  XDCC,5H  YDCC,5H  ZDCC,
3         5HTHDDC,5HPHDDC,5HPSDDC,5HXDDCC,5HYDDCC,5HZDDCC,
5         5H  AA11,5H  AA12,5H  AA13,5H  AA14,5H  AA15,5H  AA16,
6         5H  AA21,5H  AA22,5H  AA23,5H  AA24,5H  AA25,5H  AA26,
7         5H  AA31,5H  AA32,5H  AA33,5H  AA34,5H  AA35,5H  AA36,
8         5H  AA41,5H  AA42,5H  AA43,5H  AA44,5H  AA45,5H  AA46,
9         5H  AA51,5H  AA52,5H  AA53,5H  AA54,5H  AA55,5H  AA56,
A         5H  AA61,5H  AA62,5H  AA63,5H  AA64,5H  AA65,5H  AA66,
B         5H   IXX,5H   IYY,5H   IZZ,5HIMASS/
 DATA DCG/5H  AS11,5H  AS12,5H  AS13,5H  AS14,5H  AS15,5H  AS16,
1         5H  AS21,5H  AS22,5H  AS23,5H  AS24,5H  AS25,5H  AS26,
2         5H  AS31,5H  AS32,5H  AS33,5H  AS34,5H  AS35,5H  AS36,
3         5H  AS41,5H  AS42,5H  AS43,5H  AS44,5H  AS45,5H  AS46,
4         5H  AS51,5H  AS52,5H  AS53,5H  AS54,5H  AS55,5H  AS56,
5         5H  AS61,5H  AS62,5H  AS63,5H  AS64,5H  AS65,5H  AS66,
6         5H  DC11,5H  DC21,5H  DC31,5H  DC41,5H  DC51,5H  DC61,
7         5H  AR11,5H  AR21,5H  AR31,5H  AR41,5H  AR51,5H  AR61,
8         5H  AE11,5H  AE21,5H  AE31,5H  AE41,5H  AE51,5H  AE61/
 EQUIVALENCE (HANK(1),XPDDTC),(HANK(2),YPDDTC),(HANK(3),ZPDDTC)
      IF(IPRNT.EQ.1) RETURN
C
C     INITIALIZATION DONE ONLY ONCE-INTERVAL
C
      IF(JJJ.EQ.1) GO TO 5
      KKK=0
    5 CONTINUE
      KKK=KKK+1
C
C     NO. EXPERIMENTAL DATA POINTS-INPUT
C
      N=CIN(399)
```

TABLE A-1. CODE LISTING FOR THE DATACORR SUBROUTINE (Continued)

```
      WRITE(6,111) KKK,N
  111 FORMAT(/55X,8HINTERVAL,3X,I3,1H-,I3,/)
      WRITE(6,1112) TIME
 1112 FORMAT(56X,'TIME',G15.7)
C
C     COVARIANCE MATRIX,INITIAL-INPUT
C
      IF(KKK.EQ.1) GO TO 15
      DO 15 I=1,6
      DO 15 J=1,6
      AS(I,J)=SAS(I,J)
   15 CONTINUE
      IF(KKK.GT.1) GO TO 10
      DO 40 I=1,6
      DO 40 J=1,6
      AS(I,J)=0.
   40 CONTINUE
      AS(1,1)=CIN(398)
      AS(2,2)=CIN(397)
      AS(3,3)=CIN(396)
      AS(4,4)=CIN(395)
      AS(5,5)=CIN(394)
      AS(6,6)=CIN(393)
   10 CONTINUE
      JJJ=1
C
C     U70 VALUES TAKEN AS COMPUTED
C     AND EXPERIMENTAL DATA-INPUT
C
      THE=THETA
      PHE=PHI
      PSE=PSI
      THE=THE+CIN(392)
      PHE=PHE+CIN(391)
      PSE=PSE+CIN(390)
      THC=THETA
      PHC=PHI
      PSC=PSI
      THDC=THETAD
      PHDC=PHID
      PSDC=PSID
      THDDC=QDC
      PHDDC=PDC
      PSDDC=RDC
      IMASS=W
      GST=CIN( 21)
      IXX=XIXX*GST
      IYY=XIYY*GST
      IZZ=XIYY*GST
      XE=ATC+CIN(389)
      YE=BTC+CIN(388)
```

```
      ZE=GTC+CIN(387)
      XCC=ATC
      YCC=BTC
      ZCC=GTC
      XDCC=A11*VX+A12*VY+A13*VZ
      YDCC=A21*VX+A22*VY+A23*VZ
      ZDCC=A31*VX+A32*VY+A33*VZ
      XDDCC=XPDDTC
      YDDCC=YPDDTC
      ZDDCC=ZPDDTC
C
C     ERORR MATRIX
C
      AE(1,1)=THE-THC
      AE(2,1)=PHE-PHC
      AE(3,1)=PSE-PSC
      AE(4,1)=XE-XCC
      AE(5,1)=YE-YCC
      AE(6,1)=ZE-ZCC
C
C     INERTIAL TO MISSILE MATRIX-AB
C
      CTHC=COS(THC)
      STHC=SIN(THC)
      CPHC=COS(PHC)
      SPHC=SIN(PHC)
      CPSC=COS(PSC)
      SPSC=SIN(PSC)
      AB11=CPSC*CTHC-SPSC*SPHC*STHC
      AB12=CTHC*SPSC+STHC*SPHC*CPSC
      AB13=-STHC*CPHC
      AB21=-SPSC*CPHC
      AB22=CPHC*CPSC
      AB23=SPHC
      AB31=CPSC*STHC+CTHC*SPHC*SPSC
      AB32=STHC*SPSC-CTHC*SPHC*CPSC
      AB33=CTHC*CPHC
C
C     SENSITIVITY MATRIX
C
      AA(1,1)=
      AA(1,2)=
      AA(1,3)=
      AA(1,4)=
      AA(1,5)=
      AA(1,6)=
      AA(2,1)=
      AA(2,2)=
      AA(2,3)=
      AA(2,4)=
      AA(2,5)=
```

TABLE A-1. CODE LISTING FOR THE DATACORR SUBROUTINE (Continued)

```
      AA(2,6)=
      AA(3,1)=
      AA(3,2)=
      AA(3,3)=
      AA(3,4)=
      AA(3,5)=
      AA(3,6)=
      AA(4,1)=
      AA(4,2)=
      AA(4,3)=
      AA(4,4)=
      AA(4,5)=
      AA(4,6)=
      AA(5,1)=
      AA(5,2)=
      AA(5,3)=
      AA(5,4)=
      AA(5,5)=
      AA(5,6)=
      AA(6,1)=
      AA(6,2)=
      AA(6,3)=
      AA(6,4)=
      AA(6,5)=
      AA(6,6)=
C
C     MAIN GROUP PRINT ORDER
C
      PMG( 1)=THE
      PMG( 2)=PHE
      PMG( 3)=PSE
      PMG( 4)=XE
      PMG( 5)=YE
      PMG( 6)=ZE
      PMG( 7)=THC
      PMG( 8)=PHC
      PMG( 9)=PSC
      PMG(10)=XCC
      PMG(11)=YCC
      PMG(12)=ZCC
      PMG(13)=THDC
      PMG(14)=PHDC
      PMG(15)=PSDC
      PMG(16)=XDCC
      PMG(17)=YDCC
      PMG(18)=ZDCC
      PMG(19)=THDDC
      PMG(20)=PHDDC
      PMG(21)=PSDDC
      PMG(22)=XDDCC
      PMG(23)=YDDCC
```

TABLE A-1. CODE LISTING FOR THE DATACORR SUBROUTINE (Continued)

```
      PMG(24)=ZDDCC
      PMG(25)=AA(1,1)
      PMG(26)=AA(1,2)
      PMG(27)=AA(1,3)
      PMG(28)=AA(1,4)
      PMG(29)=AA(1,5)
      PMG(30)=AA(1,6)
      PMG(31)=AA(2,1)
      PMG(32)=AA(2,2)
      PMG(33)=AA(2,3)
      PMG(34)=AA(2,4)
      PMG(35)=AA(2,5)
      PMG(36)=AA(2,6)
      PMG(37)=AA(3,1)
      PMG(38)=AA(3,2)
      PMG(39)=AA(3,3)
      PMG(40)=AA(3,4)
      PMG(41)=AA(3,5)
      PMG(42)=AA(3,6)
      PMG(43)=AA(4,1)
      PMG(44)=AA(4,2)
      PMG(45)=AA(4,3)
      PMG(46)=AA(4,4)
      PMG(47)=AA(4,5)
      PMG(48)=AA(4,6)
      PMG(49)=AA(5,1)
      PMG(50)=AA(5,2)
      PMG(51)=AA(5,3)
      PMG(52)=AA(5,4)
      PMG(53)=AA(5,5)
      PMG(54)=AA(5,6)
      PMG(55)=AA(6,1)
      PMG(56)=AA(6,2)
      PMG(57)=AA(6,3)
      PMG(58)=AA(6,4)
      PMG(59)=AA(6,5)
      PMG(60)=AA(6,6)
      PMG(61)=IXX
      PMG(62)=IYY
      PMG(63)=IZZ
      PMG(64)=IMASS
C
  225 FORMAT((1H ,6(1X,A5,G15.7)))
      WRITE(6,225) (DMG(I),PMG(I),I=1,64)
C
C     COMPUTE AA TRANSPOSE=AAT
C
      IF(ICON(70).EQ.0) GO TO 30
      N=6
      NM1=N-1
      DO 30 I=1,NM1
```

TABLE A-1. CODE LISTING FOR THE DATACORR SUBROUTINE (Continued)

```
      IP1=I+1
      DO 20 J=IP1,N
      TMP1=AA(I,J)
      AA(I,J)=AAT(J,I)
   20 AAT(J,I)=TMP1
      WRITE(6,1002) AAT(3,1)
 1002 FORMAT(1X,G15.7)
   30 CONTINUE
      IF(KKK.EQ.1) GO TO 35
C
C     COMPUTE INVERSE AS =AS1
C
      DO 50 I=1,6
      DO 50 J=1,6
      X(I,J) = AS(I,J)
   50 CONTINUE
      MN1=6
      N=6
      NB=N
      MS=1
      IC=1
      ID=0
      IS=1
      CALL SESOMI(X,N,NB,MS,MN1,D,R,E,WORK,IHLD,IC,ID,IS)
      DO 60 J=1,6
      DO 60 K=1,6
      AS1(J,K)=X(J,K)
   60 CONTINUE
   35 CONTINUE
      IF(KKK.GT.1) GO TO 55
      DO 55 I=1,6
      DO 55 J=1,6
      AS1(I,J)=0.
      AS1(1,1)=1./CIN(398)
      AS1(2,2)=1./CIN(397)
      AS1(3,3)=1./CIN(396)
      AS1(4,4)=1./CIN(395)
      AS1(5,5)=1./CIN(394)
      AS1(6,6)=1./CIN(393)
   55 CONTINUE
      WRITE(14,400)
  400 FORMAT(1X,'55')
C
C     SENSITIVITY-T*COVARIANC-1 MATRIX=AL
C
      M=6
      N=6
      P=6
      DO 70 I=1,M
      DO 70 J=1,P
      AL(I,J)=0.
```

TABLE A-1. CODE LISTING FOR THE DATACORR SUBROUTINE (Continued)

```
      DO 70 K=1,N
   70 AL(I,J)=AL(I,J) + AA(K,I) * AS1(K,J)
      WRITE(14,401)
  401 FORMAT(1X,'70')
C
C     AL*SENSITIVITY MATRIX=AM
C
      M=6
      N=6
      P=6
      DO 80 I=1,M
      DO 80 J=1,P
      AM(I,J)=0.
      DO 80 K=1,N
   80 AM(I,J)=AM(I,J) + AL(I,K) * AA(K,J)
      WRITE(14,402)
  402 FORMAT(1X,'80')
C
C     AL*AE MATRIX=AN
      N=6
      P=1
      DO 90 I=1,M
      DO 90 J=1,P
      AN(I,J)=0.
      DO 90 K=1,N
   90 AN(I,J)=AN(I,J) + AL(I,K) * AE(K,J)
      WRITE(14,403)
  403 FORMAT(1X,'90')
C
C     COMPUTE INVERSE AM =AM1
C
      DO 100 I=1,6
      DO 100 J=1,6
      X(I,J)=AM(I,J)
  100 CONTINUE
      WRITE(14,404)
  404 FORMAT(1X,'100')
      MN1=6
      N=6
      NB=N
      MS=1
      IC=1
      ID=0
      IS=1
      CALL SESDMI(X,N,NB,MS,MN1,D,R,E,WORK,IHLD,IC,ID,IS)
      DO 110 J=1,6
      DO 110 K=1,6
  110 AM1(J,K) = X(J,K)
      WRITE(14,405)
  405 FORMAT(1X,'110')
C
```

TABLE A-1. CODE LISTING FOR THE DATACORR SUBROUTINE (Continued)

```
C     AM1*AN MATRIX=DC
C
      M=6
      N=6
      P=1
      DO 120 I=1,M
      DO 120 J=1,P
      DC(I,J)=0.
      DO 120 K=1,N
  120 DC(I,J)=DC(I,J) + AM1(I,K) * AN(K,J)
C
C     SENSITIVITY*DC=AP
C
      M=6
      N=M
      P=1
      DO 130 I=1,M
      DO 130 J=1,P
      AP(I,J)=0.
      DO 130 K=1,N
  130 AP(I,J)=AP(I,J) + AA(I,K) * DC(K,J)
C
C     AE-AP MATRIX=G
C
      NR=6
      NC=1
      DO 140 J=1,NC
      DO 140 I=1,NR
  140 AAR(I,J)= AE(I,J) - AP(I,J)
      N=CIN(399)
C
C     G*G -T=AS
C
      AS(1,1)=AAR(1,1)*AAR(1,1)/N
      AS(1,2)=AAR(1,1)*AAR(2,1)/N
      AS(1,3)=AAR(1,1)*AAR(3,1)/N
      AS(1,4)=AAR(1,1)*AAR(4,1)/N
      AS(1,5)=AAR(1,1)*AAR(5,1)/N
      AS(1,6)=AAR(1,1)*AAR(6,1)/N
      AS(2,1)=AAR(2,1)*AAR(1,1)/N
      AS(2,2)=AAR(2,1)*AAR(2,1)/N
      AS(2,3)=AAR(2,1)*AAR(3,1)/N
      AS(2,4)=AAR(2,1)*AAR(4,1)/N
      AS(2,5)=AAR(2,1)*AAR(5,1)/N
      AS(2,6)=AAR(2,1)*AAR(6,1)/N
      AS(3,1)=AAR(3,1)*AAR(1,1)/N
      AS(3,2)=AAR(3,1)*AAR(2,1)/N
      AS(3,3)=AAR(3,1)*AAR(3,1)/N
      AS(3,4)=AAR(3,1)*AAR(4,1)/N
      AS(3,5)=AAR(3,1)*AAR(5,1)/N
      AS(3,6)=AAR(3,1)*AAR(6,1)/N
```

```
      AS(4,1)=AAR(4,1)*AAR(1,1)/N
      AS(4,2)=AAR(4,1)*AAR(2,1)/N
      AS(4,3)=AAR(4,1)*AAR(3,1)/N
      AS(4,4)=AAR(4,1)*AAR(4,1)/N
      AS(4,5)=AAR(4,1)*AAR(5,1)/N
      AS(4,6)=AAR(4,1)*AAR(6,1)/N
      AS(5,1)=AAR(5,1)*AAR(1,1)/N
      AS(5,2)=AAR(5,1)*AAR(2,1)/N
      AS(5,3)=AAR(5,1)*AAR(3,1)/N
      AS(5,4)=AAR(5,1)*AAR(4,1)/N
      AS(5,5)=AAR(5,1)*AAR(5,1)/N
      AS(5,6)=AAR(5,1)*AAR(6,1)/N
      AS(6,1)=AAR(6,1)*AAR(1,1)/N
      AS(6,2)=AAR(6,1)*AAR(2,1)/N
      AS(6,3)=AAR(6,1)*AAR(3,1)/N
      AS(6,4)=AAR(6,1)*AAR(4,1)/N
      AS(6,5)=AAR(6,1)*AAR(5,1)/N
      AS(6,6)=AAR(6,1)*AAR(6,1)/N
  500 FORMAT(57X,17HCONVERGENCE GROUP)
C
C     RESIDUAL MATRIX-ONE ITERATION AHEAD ERROR MATRIX
C
      AR(1,1)=AAR(1,1)
      AR(2,1)=AAR(2,1)
      AR(3,1)=AAR(3,1)
      AR(4,1)=AAR(4,1)
      AR(5,1)=AAR(5,1)
      AR(6,1)=AAR(6,1)
C
C     CONVERGENCE GROUPS PRINT ORDER
C
      PCG( 1)=AS(1,1)
      PCG( 2)=AS(1,2)
      PCG( 3)=AS(1,3)
      PCG( 4)=AS(1,4)
      PCG( 5)=AS(1,5)
      PCG( 6)=AS(1,6)
      PCG( 7)=AS(2,1)
      PCG( 8)=AS(2,2)
      PCG( 9)=AS(2,3)
      PCG(10)=AS(2,4)
      PCG(11)=AS(2,5)
      PCG(12)=AS(2,6)
      PCG(13)=AS(3,1)
      PCG(14)=AS(3,2)
      PCG(15)=AS(3,3)
      PCG(16)=AS(3,4)
      PCG(17)=AS(3,5)
      PCG(18)=AS(3,6)
      PCG(19)=AS(4,1)
      PCG(20)=AS(4,2)
```

```
      PCG(21)=AS(4,3)
      PCG(22)=AS(4,4)
      PCG(23)=AS(4,5)
      PCG(24)=AS(4,6)
      PCG(25)=AS(5,1)
      PCG(26)=AS(5,2)
      PCG(27)=AS(5,3)
      PCG(28)=AS(5,4)
      PCG(29)=AS(5,5)
      PCG(30)=AS(5,6)
      PCG(31)=AS(6,1)
      PCG(32)=AS(6,2)
      PCG(33)=AS(6,3)
      PCG(34)=AS(6,4)
      PCG(35)=AS(6,5)
      PCG(36)=AS(6,6)
      PCG(37)=DC(1,1)
      PCG(38)=DC(2,1)
      PCG(39)=DC(3,1)
      PCG(40)=DC(4,1)
      PCG(41)=DC(5,1)
      PCG(42)=DC(6,1)
      PCG(43)=AR(1,1)
      PCG(44)=AR(2,1)
      PCG(45)=AR(3,1)
      PCG(46)=AR(4,1)
      PCG(47)=AR(5,1)
      PCG(48)=AR(6,1)
      PCG(49)=AE(1,1)
      PCG(50)=AE(2,1)
      PCG(51)=AE(3,1)
      PCG(52)=AE(4,1)
      PCG(53)=AE(5,1)
      PCG(54)=AE(6,1)
C
C     WRITE COVARIANC,DC,RESIDUAL,ERROR
C
      WRITE(6,500)
      WRITE(6,225) (DCG(I),PCG(I),I=1,54)
      DO 150 I=1,6
      DO 150 J=1,6
  150 SAS(I,J) = AS(I,J)
      RETURN
      END
```

APPENDIX B
MODIFICATIONS TO BOUT (Boost Output) AND RVOUT (Re-entry Vehicle Output)

APPENDIX B

MODIFICATIONS TO BOUT (Boost Output) AND RVOUT (Re-entry Vehicle Output)

The changes to the BOUT and RVOUT subroutines of the U70 simulator needed for the Tactical Ballistic Missile Trajectory Study are listed in Table B-1.

TABLE B-1. MODIFICATIONS TO BOUT AND RVOUT

PRM(70) changed to PRM (72)
RPG (162) added

```
      DIMENSION PR(30),PRE(18), PRM(72),PRG(78),RPG(162)
```

Both DO loops changed from '1,70' to '1,72'

```
      725 DO 730 IMX = 1,72
      290 CONTINUE
          IF(IMISS.EQ.0)GO TO 310
```

Added after PRM(70) = WACB

```
      PRM(71) = 0
      PRM(72) = 0

      551 CONTINUE
          PRG(22)= PRG(22)*RTOD
          PRG(23)= PRG(23)*RTOD
          PRG(28)= PRG(28)*RTOD
          PRG(29)= PRG(29)*RTOD
          PRG(34)= PRG(34)*RTOD
          PRG(35)= PRG(35)*RTOD
          PRG(41)= PRG(41)*RTOD
          PRG(64)= PRG(64)*RTOD
          PRG(65)= PRG(65)*RTOD
          PRG(75)= PRG(75)*RTOD
```

RPT(I) = PRG(I) added after PRG(75) = PRG(75)*RTOD

```
          RPG(4)=PRG(22)
          RPG(52)=PRG(23)
          RPG(10)=PRG(28)
          RPG(58)=PRG(29)
          RPG(16)=PRG(34)
          RPG(64)=PRG(35)
          RPG(137)=PRG(41)
          RPG(130)=PRG(64)
          RPG(151)=PRG(65)
          RPG(129)=PRG(75)
```

TABLE B-1. MODIFICATIONS TO BOUT AND RVOUT (Continued)

WRITE (14,26) statements added before the WRITE (6,2) statements

```
          IF(ICON(52).EQ.0) GO TO 515
          WRITE(14,25) PTIM
      515 CONTINUE
       25 FORMAT(615.7)
    C     WRITE(6,2) PTIM,TSTT,DT,IRJU,IFAIL

          IF(ICON(52).EQ.0) GO TO 516
          WRITE(14,25) TIMC
      516 CONTINUE
    C     WRITE(6,2) TIMC,TST,DT,IRJU,IFAIL

          IF(ICON(52).EQ.0) GO TO 533
          WRITE(14,26) (PR(I),I=1,30)
      533 CONTINUE
          WRITE(6,5) (DR(I),PR(I),I=1,30)

          IF(ICON(52).EQ.0) GO TO 534
          WRITE(14,26) (PRE(I),I=1,18)
      515 CONTINUE
       26 FORMAT(6(E15.7))
          WRITE(6,2) DRE(I),PRE(I),I=1,18)

          IF(ICON(52).EQ.0) GO TO 546
          WRITE(14,26) (PRM(I),I=72)
      546 CONTINUE
          WRITE(6,5) (DRM(I),PRM(I),I=1,70)

          IF(ICON(52).EQ.0) GO TO 553
          WRITE(14,26) (RPG(I),I=1,162)
      515 CONTINUE
          WRITE(6,5) (DRG(I),PRG(I),I=1,NN)
```

TABLE B-1. MODIFICATIONS TO BOUT AND RVOUT (Continued)

```
PRG(78) = TFF
IF(ICON(52).EQ.0) GO TO 220
RRG( 1) = PRG(1)
RRG( 2) = PRG(2)
RRG( 3) = PRG(3)
RRG( 4) = PRG(22)
RRG( 5) = PRG(5)
RRG( 6) = PRG(6)
RRG( 7) = PRG(7)
RRG( 8) = PRG(8)
RRG( 9) = PRG(9)
RRG(10) = PRG(28)
RRG(11) = PRG(11)
RRG(12) = PRG(12)
RRG(13) = PRG(13)
RRG(14) = PRG(14)
RRG(15) = PRG(15)
RRG(16) = PRG(34)
RRG(17) = PRG(17)
RRG(18) = PRG(18)
RRG(19) = 0
RRG(20) = 0
RRG(21) = 0
RRG(22) = 0
RRG(23) = 0
RRG(24) = 0
RRG(25) = 0
RRG(26) = 0
RRG(27) = 0
RRG(28) = 0
RRG(29) = PRG(40)
RRG(30) = 0
RRG(31) = 0
RRG(32) = 0
RRG(33) = 0
RRG(34) = 0
RRG(35) = 0
RRG(36) = 0
RRG(37) = 0
RRG(38) = 0
RRG(39) = 0
RRG(40) = 0
RRG(41) = 0
RRG(42) = 0
RRG(43) = 0
RRG(44) = 0
RRG(45) = 0
INDGC = IAPFLG + 10*IGAFLG + 100*IACFLG + 1000*ISEQ
RINDGC = INDGC + 99.D5 + 0.000005
RRG(46) = PRG(76)
```

TABLE B-1. MODIFICATIONS TO BOUT AND RVOUT (Continued)

```
      RRG(47) = 0
      RRG(48) = 0
      RRG(49) = 0
      RRG(50) = 0
      RRG(51) = 0
      RRG(52) = PRG(23)
      RRG(53) = 0
      RRG(54) = 0
      RRG(55) = 0
      RRG(56) = 0
      RRG(57) = 0
      RRG(58) = PRG(29)
      RRG(59) = PRG(53)
      RRG(60) = 0
      RRG(61) = 0
      RRG(62) = 0
      RRG(63) = 0
      RRG(64) = PRG(35)
      RRG(65) = PRG(54)
      IFTEMP = IALTU+10*(IPUDFL+1)+100*(IUPDFL+1)+1000*JBSW
     1 + 10000*(IRCHFG+1)
      RRG(66) = 0
      RRG(67) = PRG(19)
      RRG(68) = PRG(21)
      RRG(69) = 0
      RRG(70) = 0
      RRG(71) = 0
      RRG(72) = 0
      RRG(73) = PRG(25)
      RRG(74) = PRG(27)
      RRG(75) = 0
      RRG(76) = 0
      RRG(77) = 0
      RRG(78) = 0
      RRG(79) = PRG(31)
      RRG(80) = PRG(33)
      RRG(81) = 0
      RRG(82) = 0
      RRG(83) = 0
      RRG(84) = 0
      RRG(85) = 0
      RRG(86) = 0
      RRG(87) = PRG(57)
      RRG(88) = 0
      RRG(89) = 0
      RRG(90) = 0
      RRG(91) = 0
      RRG(92) = 0
      RRG(93) = PRG(63)
      RRG(94) = 0
      RRG(95) = 0
```